

PERCEIVED EXERTION AND TRAINING LOAD DURING SELF-SELECTED AND IMPOSED-INTENSITY RESISTANCE EXERCISE IN UNTRAINED WOMEN

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ABSTRACT. Focht, B.C. Perceived exertion and training load during self-selected and imposed-intensity resistance exercise in untrained women. *J. Strength Cond. Res.* 21(1):183–187. 2007.—The purpose of the present study was to examine differences in ratings of perceived exertion (RPE) and the weight lifted (training load) during self-selected and imposed-intensity bouts of acute resistance exercise (RE). Nineteen untrained college-aged women completed 2 bouts of acute resistance exercise. During 1 session, 3 sets of 4 exercises were performed using a training load of 75% of 1 repetition maximum. Conversely, during the other session, each set and exercise were completed using a self-selected training load. Assessments of RPE and training load were obtained following each set during both imposed-intensity and self-selected-intensity sessions. Results of 2 (intensity) $\times$ 3 (set) repeated measures multivariate analyses of variance revealed that, when compared to self-selected RE, RPE and resistance used were significantly higher and the number of repetitions completed per set was significantly lower during imposed-intensity RE. These findings demonstrate that the training load and perceptions of effort elicited during conventional RE prescriptions differ from the level of exertion untrained women self-select. Additionally, it appears that untrained women may not self-select a relative intensity sufficient to stimulate meaningful improvements in muscular hypertrophy or strength. The implications of these findings for the adoption and maintenance of resistance exercise participation are discussed.

KEY WORDS. effort sense, preferred exertion, strength training, percentage of 1RM

INTRODUCTION

Perceived exertion experienced during exercise can influence both the adoption and the maintenance of regular physical activity participation (6). Conventional approaches to exercise prescription advocate exercising within a specific intensity range that yields health and fitness benefits and concomitantly generates appropriate, tolerable perceptions of effort (4, 15, 16, 17). Relative exercise intensities that result in either excessive or insufficient levels of exertion may be perceived as aversive or boring and subsequently deter exercise participation. Accordingly, the extent to which commonly-employed intensity prescriptions generate perceptions of effort that are comparable to those an individual would choose to perform (i.e., self-selected intensity) is an important consideration in attempts to promote regular physical activity participation. Findings of several studies suggest that work rate and perceived exertion during self-selected-intensity bouts of aerobic exercise differ significantly from those evident during sessions completed at a prescribed intensity (7, 9). Thus, intensities imposed upon participants in traditional aerobic exercise prescriptions may not be representative of the level of exertion they select during their

own habitual exercise behavior. The discrepancy between typically imposed exercise intensities and individuals' self-selected level of exertion may contribute to the excessive attrition rates that continue to plague structured exercise programs.

Despite the growing recognition of the role of preferred exertion in the adoption and maintenance of aerobic exercise (5, 6), knowledge of intensity preferences with other modes of physical activity, such as resistance exercise, remains limited. It is well established that resistance exercise programs incorporating training loads of 70–80% or greater of an individual's 1 repetition maximum (1RM) are effective for enhancing muscular hypertrophy and strength (13, 19). However, Glass and Stanton (12) recently demonstrated that untrained men and women chose resistance exercise loads of less than 60% of 1RM. Thus, it appears that untrained individuals may self-select resistance training loads that are insufficient to induce hypertrophy or strength gains. It is also important to underscore that the selection of appropriate training intensity may be particularly problematic for untrained women who are just initiating resistance exercise participation. That is, untrained women may be more likely to be unwilling to self-select training loads adequate to produce improvements in muscular fitness because of misconceptions about resistance training, such as concern about increased risk of injury or fear of developing too much muscle mass by training with heavier loads.

At the present time, however, little is known about the training load or perceived exertion that untrained women choose to perform during self-selected-intensity resistance exercise. It is possible that the training load recommended in traditional resistance training guidelines elicits perceptions of effort that are not representative of untrained individuals' preferred workload or exertion. The exertion required to complete traditionally prescribed training protocols may be perceived as too effortful and impede continued participation. Nonetheless, it is also possible that untrained women who select too light a training load during acute resistance exercise will not experience improvements in desired aspects of muscular fitness, and the resultant lack of progress could subsequently undermine their motivation to continue resistance training. Unfortunately, no extant research has directly compared differences in training load and ratings of perceived exertion (RPE) reported during resistance exercise completed at a self-selected and imposed intensity. A previous report from this investigation focused upon the anxiolytic effects of self-selected and imposed-intensity resistance exercise (8). The results of this prior study dem-

onstrated that both self-selected and imposed-intensity resistance exercise elicited postexercise reductions in state anxiety (8). However, given the dearth of research specifically addressing untrained individuals' perceptions of effort and training load chosen during self-selected-intensity resistance exercise, the primary objective of the present study is to examine differences in RPE and training load during self-selected and imposed-intensity bouts of resistance exercise in a sample of untrained women. Based on the findings of Glass and Stanton (12) and results of prior acute aerobic research with inactive women (9), it was hypothesized that the untrained women in the present study would choose a lighter training load and report lower RPE during the self-selected-intensity bout of resistance exercise.

METHODS

Experimental Approach to the Problem

In order to examine differences in perceived exertion and training load used in imposed-intensity and self-selected-intensity bouts of acute resistance exercise, untrained college-aged women who volunteered for the study were requested to complete a 1RM testing session and 2 subsequent resistance exercise sessions. In 1 session, each participant completed 3 sets of 10 repetitions using 75% of her individual 1RM (imposed intensity), whereas in the other resistance exercise session, participants were permitted to self-select the load for each set and exercise (self-selected intensity). Ratings of perceived exertion and the training load used for each set were recorded as the outcome variables.

Subjects

Nineteen female undergraduate students (mean age = 20.6; $SD = 3.1$; 84% Caucasian) from a major university in the southeastern United States volunteered to participate in the present study. Each participant was enrolled in an introductory weight training course at the time the study was conducted. All volunteers reported regular aerobic exercise participation but had no resistance exercise experience prior to enrolling in the weight training course. Before participating, volunteers read and signed an informed consent document that had been approved by the University Institutional Review Board. Volunteers received class extra credit for participating. Based on the differences in RPE observed between self-selected and imposed-intensity exercise documented in previous research (9, 12), the sample size provided adequate statistical power to detect the main effects and interactions of interest.

Procedures

The procedures of the present investigation have been reported previously (8). However, a summary of these procedures is provided again in the following section of the present report. During the initial testing session, each subject's 1RM for the leg extension, chest press, torso-arm pull-down, and overhead press exercises was determined using a previously validated testing procedure (18). All exercises were performed using Med-X resistance training machines (Med-X, Ocala, FL). Following the completion of the 1RM session, participants completed self-selected and imposed-intensity bouts of resistance exercise as well as a quiet rest control condition. Assessments of RPE and training load were obtained only during the resistance exercise sessions. The order in which the 2 resistance exercise sessions were performed was randomly

assigned and presented in a counterbalanced fashion, and the 2 sessions were completed on separate days at least 48 hours apart. Additionally, each exercise session was completed between 10 AM and 2 PM, and the time of day each bout was performed was standardized within but not between subjects.

Three sets were completed for each exercise, and a 2-minute recovery interval was maintained between each set and exercise. Subjects were instructed to perform 10 repetitions per set, and all sets were completed to the point of volitional fatigue. During the imposed-intensity bout, all sets were performed using 75% of the participant's individual 1RM. This intensity was chosen because it is well established that it is effective for improving muscular hypertrophy and strength, and also because this resistance exercise intensity is commonly prescribed to and safe for untrained individuals (13, 19). During the self-selected intensity bout, subjects were instructed to select an appropriate resistance that would be comfortable to perform, yet still provide a good workout. Thus, in contrast to the imposed intensity session, for which a specific training load was prescribed, subjects were permitted to choose the load lifted on each set and to adjust the load at any time during the self-selected bout. Ratings of perceived exertion were assessed following the completion of each set, and the amount of resistance and training load of each set was obtained by recording the amount of resistance used and number of repetitions completed.

Measures

RPE. Overall RPE were obtained following each set during the resistance exercise sessions using the Borg 15-category scale. Subjects were provided with explicit written and verbal instructions on how to gauge their overall level of effort sense; these instructions have been described previously (3). Rating of perceived exertion has been conceptualized as a psychobiological configuration of effort sense and has been shown to be valid for use during acute resistance exercise (10, 11, 14).

Training Load. Training load during acute resistance exercise is comprised of 2 primary components: the amount of resistance lifted and the number of repetitions completed (19). Assessments of each component of training load were obtained by recording the amount of resistance and number of repetitions each subject completed for each set and exercise.

Statistical Analyses

Separate 2 (intensity: imposed and self-selected) $\times$ 3 (set: 1st, 2nd, and 3rd) repeated measures multivariate analysis of variance (MANOVA) was used to examine differences in the RPE responses, resistance used, and repetitions completed for each of the 4 resistance exercises. Univariate follow-up analyses were conducted to examine which exercises were responsible for any significant multivariate effects. Tukey post hoc analysis was employed to determine the location of significant mean differences and effect sizes (Cohen's d) accompanying selected mean changes were calculated by dividing the mean difference by the pooled standard deviation.

RESULTS

The RPE responses during the resistance exercise sessions are summarized in Table 1 and the descriptive statistics for resistance used and repetitions completed are provided in Tables 2 and 3. Participants used a resistance averaging approximately 56% of their individual 1RM

TABLE 1. Ratings of perceived exertion during self-selected and imposed-intensity resistance exercise.

Exercise	Self-selected		Imposed-intensity	
	Mean	<i>SD</i>	Mean	<i>SD</i>
Leg extension				
Set 1	12.79	1.47	15.53	1.39
Set 2	14.53	1.64	16.16	1.54
Set 3	15.68	1.97	17.05	1.47
Chest press				
Set 1	12.79	2.09	15.31	1.73
Set 2	14.63	1.92	16.26	1.75
Set 3	15.73	2.18	17.26	1.79
Torso-arm pull-down				
Set 1	11.63	2.17	14.63	1.98
Set 2	13.57	1.87	15.73	1.41
Set 3	15.21	2.09	17.11	1.66
Overhead press				
Set 1	13.74	2.07	15.31	2.36
Set 2	15.26	1.72	16.68	1.60
Set 3	16.16	1.89	17.63	1.73

during the self-selected intensity session. The MANOVA analysis of RPE responses yielded significant main effects for intensity ($p < 0.003$) and set ($p < 0.0001$). The intensity $\times$ set interaction was nonsignificant ($p < 0.13$). Univariate follow-up analyses revealed that all 4 exercises contributed to the multivariate effects for RPE. Post hoc analyses demonstrated that RPE were significantly higher ($d = 0.94$) during imposed-intensity exercise. Additionally, RPE increased significantly across sets for all exercises during both resistance exercise sessions (see Figure 1).

The MANOVA analyses of resistance used resulted in significant main effects for intensity ($p < 0.041$) and set ($p < 0.02$), as well as a significant intensity $\times$ set interaction ($p < 0.02$). Univariate analyses demonstrated that the leg extension, chest press, and torso-arm pull-down exercises contributed to the significant multivariate interaction. Post hoc analysis indicated that the significant interaction effect was attributable to the modest changes in the amount of resistance used across sets during the self-selected intensity bout (d set 1 to set 2 = 0.24; d set 1 to set 3 = 0.34) and to the absence of any change in resistance used per set during the imposed-intensity bout. Therefore, this interaction effect most likely reflects a feature of the study design that allowed subjects to alter resistance used during only the self-selected intensity bout. More importantly, however, the amount of resistance per set was significantly higher ($d = 0.93$) during the imposed-intensity bout for all exercises (see Table 2). Together, these findings indicate that subjects used significantly less resistance per set for all exercises and chose to make modest increases in the resistance across sets for the leg extension, chest press, and torso-arm pull-down exercises during self-selected intensity resistance exercise.

The MANOVA analysis of repetitions completed yielded significant main effects for intensity ($p < 0.003$) and set ($p < 0.0001$). The intensity $\times$ set interaction was nonsignificant ($p < 0.33$). Univariate follow-up analyses demonstrated that all 4 exercises contributed to the multivariate effects for repetitions completed. Post hoc analyses revealed that subjects completed more repetitions per set ($d = 1.25$) during self-selected-intensity resistance exercise. Additionally, the number of repetitions completed

TABLE 2. Resistance used during self-selected and imposed-intensity resistance exercise.

Exercise	Self-selected		Imposed-intensity	
	Mean	<i>SD</i>	Mean	<i>SD</i>
Leg extension				
Set 1	123.95	50.04	173.42	38.44
Set 2	131.05	46.74	173.42	38.44
Set 3	136.84	47.70	173.42	38.44
Chest press				
Set 1	85.26	32.42	120.00	32.27
Set 2	95.00	35.78	120.00	32.27
Set 3	93.41	36.33	120.00	32.27
Torso-arm pull-down				
Set 1	115.53	38.37	164.21	33.26
Set 2	126.58	36.71	164.21	33.26
Set 3	135.53	37.19	164.21	33.26
Overhead press				
Set 1	84.21	32.02	111.58	30.46
Set 2	87.89	30.61	111.58	30.46
Set 3	88.95	31.82	111.58	30.46

TABLE 3. Repetitions completed during self-selected and imposed-intensity resistance exercise.

Exercise	Self-selected		Imposed-intensity	
	Mean	<i>SD</i>	Mean	<i>SD</i>
Leg extension				
Set 1	9.79	0.63	8.84	1.53
Set 2	9.78	1.54	8.94	1.26
Set 3	9.47	1.12	8.53	1.47
Chest press				
Set 1	9.47	0.90	8.47	1.95
Set 2	9.47	0.96	8.05	1.71
Set 3	9.11	1.28	7.37	2.00
Torso-arm pull-down				
Set 1	9.89	0.46	9.31	1.45
Set 2	9.84	0.51	9.00	1.49
Set 3	9.79	0.63	8.26	1.66
Overhead press				
Set 1	9.58	0.77	8.10	2.31
Set 2	9.26	1.04	7.32	2.02
Set 3	8.21	1.65	56.63	1.96

decreased significantly from set 1 to set 3 for all exercises ($d = -0.47$) (see Table 3).

DISCUSSION

Recognition of the importance of intensity preferences during aerobic exercise has increased in recent years (5, 6, 9). However, this is one of the first investigations to compare perceived exertion and training load during self-selected and imposed-intensity bouts of acute resistance exercise. The present results demonstrated that, relative to imposed-intensity resistance exercise, untrained college-aged women chose to use a lighter resistance, completed more repetitions, and reported lower RPE during a bout of self-selected intensity resistance exercise. Also, consistent with findings of previous research (15, 20), RPE was found to increase across sets, whereas the number of repetitions completed decreased across sets. Taken collectively, these findings suggest that the intensity employed in conventional resistance exercise prescriptions may not be representative of the level of exertion or training load untrained women choose when permitted to self-select the intensity of activity. Women in the present in-

vestigation selected an average training load that corresponded to 56% of individual 1RM. Therefore, consistent with previous research among untrained individuals (12), the present findings also suggest that untrained women do not choose intensities sufficient to effectively produce improvements in muscular hypertrophy or strength.

Given that exertional symptoms can contribute to how favorably individuals perceive an exercise experience (6), the present findings may have important implications for the adoption and maintenance of resistance exercise. Most notably, if the resistance exercise intensity is perceived as too difficult, untrained individuals may view that activity as aversive or as too challenging and subsequently discontinue participation. Consequently, examining the affective, cognitive (i.e., self-efficacy and perceived competence), and perceptual (i.e., pain and dyspnea) responses experienced by untrained individuals during exposure to resistance exercise sessions of different intensities may aid in developing a more comprehensive understanding of the extent to which intensity and perceived exertion may influence regular resistance training participation.

The utility of using RPE to regulate intensity during aerobic exercise has been advocated previously (5). Knowledge of the value of using RPE as an adjuvant method for monitoring resistance exercise intensity remains relatively limited. Nevertheless, the present study provides some insight into this issue. For example, when given the option to choose the intensity of activity, the untrained women in this investigation selected a lighter resistance that elicited lower perceptions of exertion than those observed during the imposed-intensity session. These findings underscore the potential value of using RPE to help monitor and prescribe resistance exercise intensity. Nevertheless, it is also important to recognize that the intensity chosen during the self-selected bout (56% of 1RM) is not optimal for stimulating the improvements in muscular hypertrophy or strength often sought by resistance exercise participants. Whereas imposing an intensity that is perceived to be too effortful could undermine exercise participation, selecting a training load that is insufficient to produce improvements in muscular fitness outcomes could also deter the long-term maintenance of resistance exercise. The untrained women in the present study self-selected an intensity that was not sufficient to produce improvements in either muscular fitness (i.e., muscular strength or hypertrophy) or relevant health outcomes such as bone mineral density. Hence, in light of these issues, it is recommended that future research address the efficacy of integrating RPE into the process of prescribing resistance exercise and attempt to determine the extent to which self-regulating intensity with RPE during this mode of activity may impact adherence, the selection of training load, or the resultant health and fitness outcomes of long-term resistance training programs.

Recent resistance exercise guidelines forwarded by the American College of Sports Medicine (ACSM) suggest that a general training program involving the completion of single or multiple sets of an 8–12RM load per exercise is appropriate for untrained individuals (1). The efficacy of this protocol for producing meaningful improvements in a variety of health and fitness outcomes is well established. At the present time, however, little is known about the perceptions of exertion associated with completing these guidelines. Close inspection of the present findings suggests that the RPE responses to this common resis-

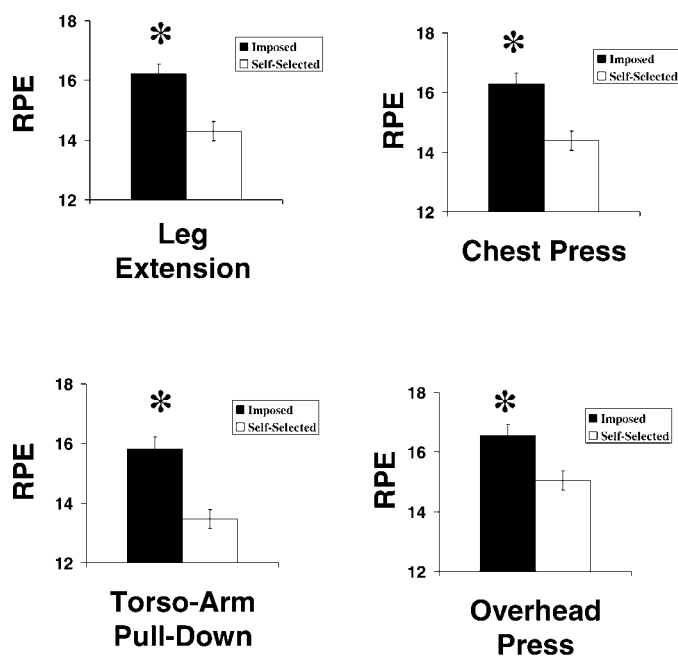


FIGURE 1. Ratings of perceived exertion (RPE) during self-selected and imposed-intensity resistance exercise.

tance exercise prescription warrant further consideration. For example, although subjects met the ACSM guidelines for repetitions on each set during both exercise sessions, RPE also increased significantly across sets during each resistance exercise protocol. Specifically, during the self-selected-intensity session, RPE increased from a rating between light and somewhat hard on set 1 to a rating exceeding hard on set 3 during the self-selected-intensity bout. During the imposed-intensity session, RPE transitioned from a rating of hard on the first set to a rating of very hard on the third set. Therefore, although multiple-set training protocols are safe and effective for untrained women, it is possible that the increased perceptions of exertion accompanying the completion of multiple sets of the same exercise could foster a sense of overexertion and potentially deter continued resistance exercise participation among novice exercisers. The link between RPE responses and adherence patterns to single- and multiple-set resistance training protocols warrants further investigation.

Arent and colleagues (2) recently demonstrated that how sets are performed has an important influence on the psychological outcomes accompanying acute resistance exercise. Consistent with this position, it should also be recognized that the criteria established for set termination may be an important consideration in determining the exertional responses during acute resistance exercise. For example, subjects in the present investigation completed each set to volitional fatigue. It is likely, however, that RPE reported during 8–12RM sets terminated at volitional fatigue are markedly different from perceptions of exertion generated during sets taken to the point of muscular failure in the same repetition range. Consequently, delineating differences in the exertional responses to sets completed to volitional fatigue and muscular failure is an important issue that should be addressed in future resistance exercise research.

Although the present study focused upon differences in total body RPE, recent findings indicate that differentiated RPE specific to active muscle groups are higher

than overall bodily perceptions of exertion during resistance exercise (14). Therefore, additional research examining preferences in active muscle RPE and how they are related to overall body perceptions of exertion during acute resistance exercise is warranted. Finally, the present sample was comprised of young women with limited resistance exercise experience. It should not be assumed that the exertion and training load chosen by this sample are representative of those which would be selected by subjects with more resistance training experience or different motivation for engaging in resistance exercise. Accordingly, future investigations in more diverse samples are necessary to extend these findings to other population subgroups, such as athletes, experienced recreational weightlifters, or individuals of other demographic subgroups (i.e., men and older adults).

In summary, the primary finding of this study indicates that training load and RPE responses are significantly different during acute bouts of resistance exercise performed at an imposed and self-selected intensity among untrained women. Exertional preferences may impact habitual resistance exercise participation, and this relationship warrants further investigation.

PRACTICAL APPLICATIONS

Given that the results of the present study suggest that untrained women do not select a training load sufficient to induce hypertrophy or strength gains, these findings have a number of important implications. First, it is essential that fitness professionals recognize that untrained individuals may not select appropriate intensities to stimulate desired changes in muscular strength and size. Lack of progress because of inappropriate intensity selection may undermine clients' motivation to continue with the resistance training program and subsequently lead to program attrition. Nevertheless, it is possible that some untrained participants may perceive traditional resistance training prescriptions of 70–80% of 1RM or 6–12 RM as too effortful. A discrepancy between preferred level of exertion and the effort sense accompanying such imposed-intensity resistance exercise sessions may increase the likelihood of dropping out. Therefore, it is essential that fitness professionals closely monitor intensity and perceived exertion during the adoption phase of resistance exercise programs in inexperienced weight lifters. Implementing sensible progression in training that is tailored to the individual's exercise abilities and exertional preferences can help to optimize the untrained client's improvements in muscular strength and size without compromising the client's motivation to continue the training program. It is also important that fitness professionals help educate untrained women about resistance training misconceptions that may prevent them from using the appropriate training loads necessary to elicit improvements in hypertrophy and strength. Many women may select lighter training loads because of fears of becoming bulky or too muscular, and such concerns can prevent individuals from achieving progress towards their

fitness goals. Finally, trainers can utilize perceived exertion to help untrained individuals more effectively identify and self-regulate appropriate resistance exercise intensity. Incorporating RPE in the prescription of resistance training could facilitate progression and adherence among novice exercisers.

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